

Title : Sparse matrix operations

Problem Stat Statement:

- write a C program for sparse matrix realization and operations on it - Transpose fast transpose

Objective .

- 1) To study the concept of sparse matrix, how it is stored and displayed
- 2) To understand the implementation of sparse matrix operations - simple transpose, fast transpose.

Implementation

Platform:

- 64-bit open source linux on it's derivatives
- Open source C programming tool like gcc / Eclipse Editor
- Pseudo code
matrix to compact formatting

Procedure Convert to triplet (matrix [row][col])

triplet [Max][3]

Initialize triplet [0][0] = row

Initialize triplet [0][1] = col

Initialize k = 1;

for $i = 0$ to, row $= 1$;

for $i = 0$ to col $- 1$ do;

if matrix $[i][j] \neq 0$ then;
triplet $[k][0] = i$
triplet $[k][1] = j$
triplet $[k][2] = \text{matrix}[i][j]$
 $k = k + 1$

End if

End for

End for

Triplet $[0][2] = k - 1$

End procedure

Simple transpose:

Procedure Simple transpose $[n][max][C3]$, $TC[Max][C3]$

$T[0][0] = M[0][1]$

$T[0][1] = M[0][0]$

$T[0][2] = M[0][2]$

Initialize $k = 1$

Num Lods $= M[0][1]$

Num term $= M[0][2]$

For col $= 0$ to Num L - 1 do;

For $i = 1$ to num terms do;

if $M[i][1] = \text{col}$ then

$T[k][0] = M[i][1]$

$T[k][1] = M[i][0]$

$T[k][C_2] = M[C_1][C_2]$
 $k = k + 1$

End if
 End for
 End for

End procedure

→ Fast transpose

→ Procedure Fast Trans ($M[A \times B][1]$, $T[Max][3]$)

$T[0][0] = M[0][0]$
 $T[0][1] = M[0][0]$
 $T[0][2] = M[0][0]$

Initialize Row terms [Num cols] with 0
 Initialize startypes [Num cols] with 0

for $i = 1$ to num terms do
 Row terms [$M[C_1][C_1]$] = Row terms [$M[C_1][C_1]$]
 End for

~~Start~~ Startypes [0] = 1
 for $k = 1$ to Num cols - 1 do
 startypes [C_1] = startypes [$C_1 - 1$] + [$C_1 - 1$]
 End for

for $i = 1$ to Num terms do
 col = $M[C_1][C_1]$
 for = startypes [col]

$T[pos][0] = M[C_1][C_1]$
 $+ [pos][1] = M[C_1][0]$
 $T[pos][2] = M[C_1][C_2]$

start types Col D = start start types Col + 1
End for
End procedure

line complexity.

C = columns in matrix, N = no of element

for simple transpose $\rightarrow O(L \times N)$

for fast Transpose $\rightarrow O(N \log C)$

FAQs

Q What is a sparse matrix? list two applications

A sparse matrix has a majority of element as zero, storing there is a standard 2D array is a waste of space and time. They are majorly applied in ML & NLP: Term doc matrices.

Q Represent sparse matrix with suitable data structure? Explain with an example simple & fast transpose!

$\Rightarrow$ Sparse matrix can be represented using triplet representation. It is usually a 2-D array with three columns that represent the location and value of the non-zeros.

Q) Find of the addition of two sparse matrices in triplet form also find single and post transposing.

Ans let 2 triplet form of matrices is:

$$M_1 = \begin{bmatrix} 4 & 5 & 6 \\ 0 & 3 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 45 \\ 2 & 3 & 4 \\ 3 & 2 & 45 \\ 4 & 1 & 2 \end{bmatrix} \quad M_2 = \begin{bmatrix} 4 & 5 & 5 \\ 0 & 3 & 7 \\ 0 & 4 & 6 \\ 1 & 4 & 4 \\ 2 & 1 & 8 \\ 3 & 2 & 45 \\ 4 & 4 & 21 \end{bmatrix}$$

Addition: we add the matching indices values, and if the indices don't match it gets added as a new row in the result. In the result value result

$$\begin{aligned} (0, 3) &\rightarrow 5 + 7 = 12 \\ (1, 4) &\rightarrow 45 + 4 = 49 \\ (3, 2) &\rightarrow 45 + 45 = 90 \end{aligned}$$

No match

$$\begin{aligned} (0, 4) &\rightarrow 6 \\ (1, 3) &\rightarrow 8 \\ (2, 1) &\rightarrow 8 \\ (2, 3) &\rightarrow 4 \\ (4, 1) &\rightarrow 2 \\ (4, 4) &\rightarrow 21 \end{aligned}$$

These were arranged acc. to indices here

$$M_1 + M_2 = \begin{bmatrix} 4 & 5 & 9 \\ 0 & 3 & 12 \\ 0 & 4 & 6 \\ 1 & 3 & 8 \\ 1 & 4 & 49 \\ 2 & 1 & 8 \\ 2 & 3 & 4 \\ 3 & 2 & 90 \\ 4 & 2 & 2 \\ 4 & 4 & 21 \end{bmatrix}$$

single or post transpose have different
but the same result.

Transpose of the resulting matrix.

$$\begin{bmatrix} 5 & 4 & 9 \\ 1 & 2 & 8 \\ 1 & 4 & 2 \\ 2 & 3 & 40 \\ 3 & 0 & 12 \\ 3 & 1 & 8 \\ 3 & 2 & 7 \\ 4 & 0 & 6 \\ 4 & 1 & 48 \\ 4 & 4 & 22 \end{bmatrix}$$

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